

GLOBALFOUNDRIES® 22FDX®
GF22FDX_SC8T_104CPP_BASE_CSC28LVT
Standard Cells Library

Release Notes, Known Issues, and Recommendations
Version: 6.40

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Revision History

A list of changes appearing in the document.

Version	Date	Changes
6.40	December 12, 2019	PGV support is enabled for static bias and dynamic bias PVTs along with zero bias PVTs
6.30	September 30, 2019	• PVTs are added • Fixed the large OCV constraints issue • PGV views are added
6.20	May 31, 2019	APL and Tetramax views are updated
6.10	March 15, 2019	Added BIAS PVTs
6.00	November 12, 2018	This release is based on PDK 1.3_2.2
5.00	July 28, 2017	PDK Update
4.10	May 22, 2017	PDK Update
4.00	May 2, 2017	PDK Update
3.00	December 14, 2016	PDK Update
2.20	July 29, 2016	Fifth Release (Updated TAP cell and Verilog view)
2.10	July 5, 2016	Fourth Release (Updated Milkyway, clf and edge_constraints files)
2.00	June 22, 2016	Third Release (PDK Update)
1.01	May 23, 2016	Second Release (Modified ANTENNA cell)
1.00	April 15, 2016	First Release

1 About this Document

This is a design kit release for GF22FDX_SC8T_104CPP_BASE_CSC28LVT with (8 Track Standard Cell LVT) 959 cells. For the supported PVTs in the product, refer to *doc/CORNERS.txt*. For complete cell list, refer to *doc/GF22FDX_SC8T_104CPP_BASE_CSC28LVT.cell_list*.

1.1 Intended Audience

This document is intended to assist the design engineers in the use of INVECAS® Standard Cells Library products for System-on-Chip (SoC) or in general any chip development.

1.2 Document Organization

This document is organized into the following sections:

1. [Release Notes](#)
2. [Known Issues](#)
3. [Recommendations](#)
4. [Appendix: Supported Views](#)

2 Release Notes

2.1 Changes in Version 6.40 with respect to 6.30

- As per the Change Notice # CN019, PGV views for adaptive and static bias were removed in the previous versions of the libraries. PGV support is now enabled for adaptive and static bias PVTs along with zero bias PVTs.

2.2 Changes in Version 6.30 with respect to 6.20

- The following PVTs are added:
 - SSG_OP59V_OP00V_OP60V_M0P80V_125C
 - SSG_OP59V_OP00V_OP85V_M1P50V_M40C
 - SSG_OP72V_OP00V_OP60V_M1P00V_125C
 - SSG_OP72V_OP00V_OP70V_M1P50V_M40C
 - SSG_OP81V_OP00V_OP60V_M0P90V_125C
 - SSG_OP81V_OP00V_OP60V_M1P50V_M40C
- Fixed the higher OCV numbers for constraints in liberty files
- PGV view support is enabled

2.3 Changes in Version 6.20 with respect to 6.10

- Tetramax views are updated for sequential cells to resolve port-net connectivity issue while doing DFT analysis.
- APL views are re-generated after updating surround context for LPE extraction

2.4 Changes in Version 6.10 with respect to 6.00

- The following PVTs are added:
 - FFG_OP72V_OP00V_OP45V_M0P45V_125C
 - FFG_OP72V_OP00V_OP45V_M0P45V_M40C
 - FFG_OP72V_OP00V_OP80V_M0P80V_125C
 - FFG_OP72V_OP00V_OP80V_M0P80V_M40C
 - FFG_OP88V_OP00V_OP45V_M0P45V_125C
 - FFG_OP88V_OP00V_OP45V_M0P45V_M40C
 - FFG_OP88V_OP00V_OP80V_M0P80V_125C
 - FFG_OP88V_OP00V_OP80V_M0P80V_M40C
 - FFG_OP945V_OP00V_OP50V_M0P50V_125C
 - FFG_OP945V_OP00V_OP50V_M0P50V_M40C
 - FFG_OP945V_OP00V_OP90V_M0P90V_125C
 - FFG_OP945V_OP00V_OP90V_M0P90V_M40C
 - SSG_OP59V_OP00V_OP45V_M0P45V_125C
 - SSG_OP59V_OP00V_OP45V_M0P45V_M40C
 - SSG_OP59V_OP00V_OP80V_M0P80V_125C
 - SSG_OP59V_OP00V_OP80V_M0P80V_M40C
 - SSG_OP72V_OP00V_OP45V_M0P45V_125C
 - SSG_OP72V_OP00V_OP45V_M0P45V_M40C
 - SSG_OP72V_OP00V_OP80V_M0P80V_125C
 - SSG_OP72V_OP00V_OP80V_M0P80V_M40C
 - SSG_OP76V_OP00V_OP45V_M0P45V_125C
 - SSG_OP76V_OP00V_OP45V_M0P45V_M40C
 - SSG_OP76V_OP00V_OP80V_M0P80V_125C
 - SSG_OP76V_OP00V_OP80V_M0P80V_M40C
 - SSG_OP81V_OP00V_OP50V_M0P50V_125C
 - SSG_OP81V_OP00V_OP50V_M0P50V_M40C

- SSG_OP81V_OP00V_OP90V_M0P90V_125C
- SSG_OP81V_OP00V_OP90V_M0P90V_M40C
- TT_OP59V_OP00V_OP45V_M0P45V_125C
- TT_OP59V_OP00V_OP45V_M0P45V_M40C
- TT_OP65V_OP00V_OP45V_M0P45V_25C
- TT_OP65V_OP00V_OP45V_M0P45V_85C
- TT_OP65V_OP00V_OP80V_M0P80V_25C
- TT_OP65V_OP00V_OP80V_M0P80V_85C
- TT_OP72V_OP00V_OP45V_M0P45V_125C
- TT_OP72V_OP00V_OP45V_M0P45V_M40C
- TT_OP80V_OP00V_OP45V_M0P45V_25C
- TT_OP80V_OP00V_OP45V_M0P45V_85C
- TT_OP80V_OP00V_OP80V_M0P80V_25C
- TT_OP80V_OP00V_OP80V_M0P80V_85C
- TT_OP81V_OP00V_OP50V_M0P50V_125C
- TT_OP81V_OP00V_OP50V_M0P50V_M40C
- TT_OP90V_OP00V_OP50V_M0P50V_25C
- TT_OP90V_OP00V_OP50V_M0P50V_85C
- TT_OP90V_OP00V_OP90V_M0P90V_25C
- TT_OP90V_OP00V_OP90V_M0P90V_85C
- PGV Support has been disabled because of Cadence flow issue

2.5 Changes in Version 6.00 with respect to 5.00

- The following cells are newly added:
 - SC8T_AOI22X3_CSC*
 - SC8T_BUFX24_CSC*
 - SC8T_CKBUX24_CSC*
 - SC8T_CKGPRELATNX12_CSC*
 - SC8T_CKGPRELATNX16_CSC*
 - SC8T_CKGPRELATNX24_CSC*
 - SC8T_CKGPRELATNX8_CSC*
 - SC8T_CKGPRELATX12_CSC*
 - SC8T_CKGPRELATX16_CSC*
 - SC8T_CKGPRELATX1_CSC*
 - SC8T_CKGPRELATX24_CSC*
 - SC8T_CKGPRELATX2_CSC*
 - SC8T_CKGPRELATX4_CSC*
 - SC8T_CKGPRELATX8_CSC*
 - SC8T_CKINVX24_CSC*
 - SC8T_INVX24_CSC*
 - SC8T_INVX3_CSC*
 - SC8T_LATNRQX1_CSC*
 - SC8T_LATNRQX2_CSC*
 - SC8T_LATNRQX4_CSC*
 - SC8T_LATRQX1_CSC*
 - SC8T_LATRQX2_CSC*
 - SC8T_LATRQX4_CSC*
 - SC8T_ND2X3_CSC*
 - SC8T_NR2X3_CSC*
 - SC8T_OAI22X3_CSC*
 - SC8T_AN3IAX0P5_CSC*
 - SC8T_AN3IAX1_CSC*
 - SC8T_AN3IAX2_CSC*
 - SC8T_AN3IAX4_CSC*

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- SC8T_AN3IAX6_CSC*
 - SC8T_AN4IAX0P5_CSC*
 - SC8T_AN4IAX1_CSC*
 - SC8T_AN4IAX2_CSC*
 - SC8T_AN4IAX4_CSC*
 - SC8T_AOA211X0P5_CSC*
 - SC8T_AOA211X1_CSC*
 - SC8T_AOA211X2_CSC*
 - SC8T_AOA211X4_CSC*
 - SC8T_AOA211X6_CSC*
 - SC8T_NR3IBX0P5_CSC*
 - SC8T_NR3IBX1_CSC*
 - SC8T_NR3IBX2_CSC*
 - SC8T_NR3IBX4_CSC*
 - SC8T_NR3IBX6_CSC*
 - SC8T_OR2IAMUX2X0P5_CSC*
 - SC8T_OR2IAMUX2X1_CSC*
 - SC8T_OR2IAMUX2X2_CSC*
 - SC8T_OR2IAMUX2X4_CSC*
 - SC8T_OR2IAMUX2X6_CSC*
 - SC8T_OR4IAX0P5_CSC*
 - SC8T_OR4IAX1_CSC*
 - SC8T_OR4IAX2_CSC*
 - SC8T_OR4IAX4_CSC*
 - SC8T_OR4IAX6_CSC*
 - SC8T_OR6X0P5_CSC*
 - SC8T_OR6X1_CSC*
 - SC8T_OR6X2_CSC*
 - SC8T_OR6X4_CSC*
 - SC8T_OR6X6_CSC*
 - The following cells are updated for increasing PMOS RX enclosure of CA to 20 nm:
 - SC8T_AO211X1P5_CSC*
 - SC8T_AO21X1P5_CSC*
 - SC8T_AO221X0P5_CSC*
 - SC8T_AO221X1_MR_CSC*
 - SC8T_AO31X1P5_CSC*
 - SC8T_AOI211X0P5_CSC*
 - SC8T_DFFNX2_CSC*
 - SC8T_DFFNX4_CSC*
 - SC8T_DFFQX4_CSC*
 - SC8T_DFFRQNX1_CSC*
 - SC8T_DFFRQX1_CSC*
 - SC8T_DFFX2_CSC*
 - SC8T_LATQX2_CSC*
 - SC8T_ND4IABX0P5_CSC*
 - SC8T_ND4IABX1_CSC*
 - SC8T_ND4IABX1_MR_CSC*
 - SC8T_NR3IAX1P5_CSC*
 - SC8T_NR4IABX0P5_CSC*
 - SC8T_NR4IABX1_CSC*
 - SC8T_NR4IABX1_MR_CSC*
 - SC8T_OA211X1P5_CSC*
 - SC8T_OAI21IAX1_CSC*
 - SC8T_OAI21IAX1_MR_CSC*
 - SC8T_OAI311X0P5_CSC*

- SC8T_OAI31X1P5_CSC*
- SC8T_OR2X1P5_CSC*
- SC8T_SDIFFNX2_CSC*
- SC8T_SDIFFNX4_CSC*
- SC8T_SDIFFRQX0P5_CSC*
- SC8T_SDIFFRQX1_CSC*
- SC8T_SDIFFRQX2_CSC*
- SC8T_SDIFFSQNX1_CSC*
- Updated DECAP and FILL cells to fix density issues:
 - SC8T_DECAPX16_CSC*
 - SC8T_DECAPX4_CSC*
 - SC8T_DECAPX5_CSC*
 - SC8T_DECAPX8_CSC*
 - SC8T_FILLX128_CSC*
 - SC8T_FILLX16_CSC*
 - SC8T_FILLX32_CSC*
 - SC8T_FILLX64_CSC*
- Added PIN Connectivity information in GDS and CDSLIB
- Updated Site name, Antenna gate area value, added PROPERTY DEFINITIONS, and LEF58_CLASS property in LEF
- The following cells are optimized for Pin Access:
 - SC8T_AO22IA1A2X6_CSC*
 - SC8T_AO22IA1A2X6_MR_CSC*
 - SC8T_CKINVX12_CSC*
 - SC8T_DFFX2_CSC*
 - SC8T_MUXI2X6_CSC*
 - SC8T_MUXT2X0P5_CSC*
 - SC8T_MUXT2X1_CSC*
 - SC8T_MUXT2X1_MR_CSC*
 - SC8T_OAI222X1_CSC*
 - SC8T_OAI222X0P5_CSC*
 - SC8T_OAI222X1_MR_CSC*
 - SC8T_AO32X0P5_CSC*
 - SC8T_AO32X1_CSC*
 - SC8T_AO32X1_MR_CSC*
 - SC8T_OA32X0P5_CSC*
 - SC8T_OA32X1_CSC*
 - SC8T_OA32X1_MR_CSC*
 - SC8T_AOI33X1_CSC*
 - SC8T_AOI33X1_MR_CSC*
 - SC8T_AOI33X0P5_CSC*
 - SC8T_OAI33X1_CSC*
 - SC8T_OAI33X0P5_CSC*
 - SC8T_OAI33X1_MR_CSC*
 - SC8T_AO33X1_CSC*
 - SC8T_AO33X0P5_CSC*
 - SC8T_AO33X1_MR_CSC*
 - SC8T_NR4IABX2_CSC*
- All the CK (clock) cells are updated for Clock Balancing

2.6 Changes in Version 5.00 with respect to 4.10

- The library database is based on the pdk-22FDX-V1.2_0.0.
- Approximately 25 cells have been modified in order to align with DRC rules based on pdk-22FDX-V1.2_0.0.

2.7 Changes in Version 4.10 with respect to 4.00

- Tetramax view has been provided
- PGV has been provided for Fast and Nominal PVT corners
- Following new PVT has been added
 - FFG_0P99V_0P00V_0P00V_0P00V_125C
 - FFG_0P99V_0P00V_0P00V_0P00V_M40C
 - SSG_0P81V_0P00V_0P00V_0P00V_125C
 - SSG_0P81V_0P00V_0P00V_0P00V_M40C
 - TT_0P90V_0P00V_0P00V_0P00V_85C
 - TT_0P90V_0P00V_0P00V_0P00V_25C
- Liberty files are supported in .gz format

2.8 Changes in Version 4.00 with respect to 3.00

- The library database contains 899 cells
- The library database is based on the pdk-22FDSOI_V1.1_0.0
- Cells have been modified in order to align with DRC/DFM rules based on pdk-22FDSOI_V1.1_0.0

2.9 Changes in Version 3.00 with respect to 2.20

- Library has been updated for DRCs based on pdk-22FDSOI_V1.0_0.0
- Added 701 cells to the library
- Following cells are renamed
 - SC8T_CNREXTANTENNANLX10_CSC28L => SC8T_CNREXTANTENNANLX9_CSC28L
 - SC8T_CNREXTANTENNANRX10_CSC28L => SC8T_CNREXTANTENNANRX9_CSC28L
 - SC8T_CNREXTANTENNAPLX10_CSC28L => SC8T_CNREXTANTENNAPLX9_CSC28L
 - SC8T_CNREXTANTENNAPRX10_CSC28L => SC8T_CNREXTANTENNAPRX9_CSC28L
 - SC8T_CONCAVEANTENNANLX10_CSC28L => SC8T_CONCAVENLX9_CSC28L
 - SC8T_CONCAVEANTENNANRX10_CSC28L => SC8T_CONCAVENRX9_CSC28L
 - SC8T_CONCAVEANTENNAPLX10_CSC28L => SC8T_CONCAVEPLX9_CSC28L
 - SC8T_CONCAVEANTENNAPRX10_CSC28L => SC8T_CONCAVEPRX9_CSC28L
 - SC8T_ROWCAPANTENNANLX10_CSC28L => SC8T_ROWCAPANTENNANLX9_CSC28L
 - SC8T_ROWCAPANTENNANRX10_CSC28L => SC8T_ROWCAPANTENNANRX9_CSC28L
 - SC8T_CNREXTANTENNANLX12_CSC28L => SC8T_CNREXTANTENNANLX11_CSC28L
 - SC8T_CNREXTANTENNANRX12_CSC28L => SC8T_CNREXTANTENNANRX11_CSC28L
 - SC8T_CNREXTANTENNAPLX12_CSC28L => SC8T_CNREXTANTENNAPLX11_CSC28L
 - SC8T_CNREXTANTENNAPRX12_CSC28L => SC8T_CNREXTANTENNAPRX11_CSC28L
 - SC8T_CONCAVEANTENNANLX12_CSC28L => SC8T_CONCAVENLX11_CSC28L
 - SC8T_CONCAVEANTENNANRX12_CSC28L => SC8T_CONCAVENRX11_CSC28L
 - SC8T_CONCAVEANTENNAPLX12_CSC28L => SC8T_CONCAVEPLX11_CSC28L
 - SC8T_CONCAVEANTENNAPRX12_CSC28L => SC8T_CONCAVEPRX11_CSC28L
 - SC8T_ROWCAPANTENNANLX12_CSC28L => SC8T_ROWCAPANTENNANLX11_CSC28L
 - SC8T_ROWCAPANTENNANRX12_CSC28L => SC8T_ROWCAPANTENNANRX11_CSC28L

2.10 Changes in Version 2.20 with respect to 2.10

- Modified TAP cell to meet High voltage DRCs
- Verilog models are updated to support negative timing

2.11 Changes in Version 2.10 with respect to 2.00

- Updated Milkyway, clf and edge constraints files

2.12 Changes in Version 2.00 with respect to 1.01

- Library has been updated for DRCs and DFM based on pdk-22FDSOI_V0.5_0.0
- Complete re-characterization done with pdk-22FDSOI_V0.5_0.0
- Following cells are renamed

• SC8T_ANTENNAX1_CSC*	=> SC8T_ANTENNAX11_CSC*
• SC8T_CNREXTANTENNANLX9_CSC*	=> SC8T_CNREXTANTENNANLX10_CSC*
• SC8T_CNREXTANTENNANRX9_CSC*	=> SC8T_CNREXTANTENNANRX10_CSC*
• SC8T_CNREXTANTENNAPLX9_CSC*	=> SC8T_CNREXTANTENNAPLX10_CSC*
• SC8T_CNREXTANTENNAPRX9_CSC*	=> SC8T_CNREXTANTENNAPRX10_CSC*
• SC8T_CONCAVEANTENNANLX9_CSC*	=> SC8T_CONCAVENLX10_CSC*
• SC8T_CONCAVEANTENNANRX9_CSC*	=> SC8T_CONCAVENRX10_CSC*
• SC8T_CONCAVEANTENNAPLX9_CSC*	=> SC8T_CONCAVEPLX10_CSC*
• SC8T_CONCAVEANTENNAPRX9_CSC*	=> SC8T_CONCAVEPRX10_CSC*
• SC8T_ROWCAPANTENNANLX9_CSC*	=> SC8T_ROWCAPANTENNANLX10_CSC*
• SC8T_ROWCAPANTENNANRX9_CSC*	=> SC8T_ROWCAPANTENNANRX10_CSC*
• SC8T_TAPX7_CSC*	=> SC8T_TAPX8_CSC*
• SC8T_TAPZBX7_CSC*	=> SC8T_TAPZBX8_CSC*
- Following cells are added
 - SC8T_FILLX1_CSC*

2.13 Changes in Version 1.01 with respect to 1.00

- SC8T_ANTENNAX1_CSC28L cell is modified with adding one diode between VPW_P and VSS

2.14 Initial Release (1.00)

- This is initial release of the library
- The library database contains 132 cells (108 Logical and 24 Physical cells)
- The library database is based on the pdk-22FDSOI_V0.4_1.0
 - Power Antenna diode is added for the following cells.
 - SC8T_CNREXTANTENNANLX9_CSC28L
 - SC8T_CNREXTANTENNANRX9_CSC28L
 - SC8T_CNREXTANTENNAPLX9_CSC28L
 - SC8T_CNREXTANTENNAPRX9_CSC28L
 - SC8T_ROWCAPANTENNANLX9_CSC28L
 - SC8T_ROWCAPANTENNANRX9_CSC28L
 - No M1 column cap version of cells added, to avoid NWEELL Antenna effect for bigger designs.
 - SC8T_NOM1COLCAPNX1_CSC28L
 - SC8T_NOM1COLCAPPX1_CSC28L
- Supports mixed-VT abutment (LVT-SLVT) and (RVT-HVT) independently
- This product is a continuation of GF22fdsoi_sc8_104cpp_pcnrx_c28lvt version 2016q1v1. Product name is modified from GF22fdsoi_sc8_104cpp_pcnrx_c28lvt to GF22FDX_SC8T_104CPP_BASE_CSC28L. All cell names are aligned with product name

3 Known Issues

The following are the known issues in GF22FDX_SC8T_104CPP_BASE_CSC28LVT library:

1. 'GRCLC.OUTLINE.C.2_M1_U_cfill_all' is the cell level DRC violation based on PDK V1.3_2.2, which will be cleaned at design level.
2. The following clock cells should not be placed under power mesh:
 - SC8T_CKNR2X16_CSC*
 - SC8T_CKND2X16_CSC*
3. Standard cell libraries are designed to follow certain performance trends (including variations) across the drives and PVTs. However, there may be cases, depending on cell architecture and simulation conditions, where trends may not follow the expected lines.
4. Intrinsic capacitance numbers present in CCS Liberty files are pessimistic, because of Cadence® Liberate tool limitation. The issue is under debug with Cadence.

4 Recommendations

The following are the recommendations for using GF22FDX_SC8T_104CPP_BASE_CSC28LVT library.

1. For more information on supported views refer to [Appendix: Supported Views](#), and for supported PVTs, see *doc/CORNERS.txt*.
2. Refer to *doc/GF22FDX_SC8T_104CPP_BASE_CSC28L_decap.xlsx* for decap capacitance values.
3. Ignore Warning/Error messages related to ANTENNADIFFAREA that might appear for the output pins while loading in Cadence INNOVUS tool.
4. 'GRPC.S.14.or' DRC will occur if the following placement guidelines for placing the TAP and ANTENNA cells are not followed:
 - Between any two TAP cells, vertical abutment should be exact without any offset.
 - Similarly for any two antenna cells, vertical abutment should be exact without any offset
 - Avoid vertical abutment combination between Antenna and TAP cells.
5. For designs with vertical poly, set below variable to "FALSE".
 - setenv PLORIENT_PREFER_HORIZ
6. Cell EDGE properties, that is SOURCE/DRAIN information, are provided in the edge_constraints file (*scripts/milkyway/edge_constraints_*.tcl*) present under the scripts directory and are useful for SYNOPSIS-based ICC flow.
7. Cadence and Synopsys-based EDA flows support 1CPP-aware placement, and 1CPP placement gap allows only between DRAIN-DRAIN combination. It should be prohibited in rest of the placement combinations.
8. Standard cell libraries are designed to meet EM guidelines for normal chip/IP design usage. However, it is the responsibility of the chip/IP designer to ensure that EM is met at the chip/IP level with regard to specific guidelines/recommendations.
9. Refer to the hcell list present in the *scripts/lvs* directory for running hierarchical LVS at design level.
10. Cell name conflicts must be resolved at top-level.
11. Please follow the below guidelines for core Diode cell placement:
 - Placement of CORE Diode cells should be in a vertical stack pattern (No checkered pattern) in order not to get any DRC violations.
 - Between any two CORE Diode cells, vertical abutment should be exact without any offset.
 - These cells should be terminated at the core boundaries, that is top and bottom rows, with their corresponding *COLCAPDIODE* cells.
 - Placement of TAP cells is mandatory in between the CORE Diode cells and ROW termination cells (*ROWCAP*), that is when these cells are separated horizontally.
12. Refer to the placement settings script present in the *scripts/icc* directory for running Synopsys' ICC-based design implementation and *scripts/icc2* directory for running Synopsys' ICC2-based design implementation.
13. Include the following Physical cells while writing out the PG Verilog as the cell layout contains diode region:
 - SC8T_ROWCAPANTENNA*
 - SC8T_CNREXTANTENNA*
 - SC8T_DIODE*
14. Placement spacing labels and their corresponding spacing rules are defined in the Standard Cell lef file. By default, Cadence-based EDA flow honors these placement constraints and places the cells based on the spacing rules that are defined. There is no specific action required from the designer.
15. Refer to the placement spacing labels and constraints, which are present in the scripts of icc/icc2 directory for Synopsys' ICC/ICC2-based design implementation. Contact INVECAS/GF for any further assistance.
16. Intrinsic capacitance database should not be referred for design level IR Drop Analysis.

17. The following table provides list of Standalone DRC Violations.

Table 1 : Standalone DRC Violations

Violations	
GRAUX3.C.2	GRNPLUS.PPLUS.C.1
GRAUX3.C.9	GRNPLUS.S.6
GRBOOL.3PL.W.2	GRNPLUS.S.7
GRBOOL.BF.A.1	GRNPLUS.W.1
GRBOOL.LW.A.1	GRNPLUS.W.1a
GRBOOL.MY.A.1	GRNW.A.1
GRBOOL.N3.A.1	GRNW.D.FW.1
GRBOOL.N3.W.1	GRNW.D.FW.2
GRBOOL.NW.A.1	GRNW.D.HYB.2
GRCLC.OUTLINE.C.2_M1_U_cfill_all	GRNW.EN.RX.1
GRCLC.OUTLINE.C.2_pc_no_cut	GRNW.EX.GT.1
GRcRX.C.5	GRNW.W.1
GRCT.C.3	GRPC.C.8
GRCT.EX.PC.2	GRPC.EN.1
GRCT.EX.PC.3	GRPC.S.14.or
GRCT.OL.PC.1	GRPC.W.5
GRDIODE.A.1	GRPPLUS.A.1
GRDIODE.C.4_DIODE_keepout_out	GRPPLUS.C.6
GRDIODE.W.1	GRPPLUS.C.7
GRGT.S.1.or	GRPPLUS.C.8
GRHYBRID.A.1	GRPPLUS.EN.RX.1
GRHYBRID.C.2	GRPPLUS.EX.GT.1
GRHYBRID.C.5	GRPPLUS.S.6
GRHYBRID.W.1	GRPPLUS.S.7
GRLVTN.A.1	GRPPLUS.W.1
GRLVTN.W.2	GRPPLUS.W.1a
GRLVTP.A.1	GRS3.A.1
GRLVTP.W.2	GRS3.EN.NW.1
GRNPLUS.A.1	GRS3.S.2
GRNPLUS.C.6	GRS3.W.1
GRNPLUS.C.9	GRxVTP.EN.GT.1
GRNPLUS.EN.RX.1	GRxVTP.EN.RX.1
GRNPLUS.EX.GT.1	—

5 Appendix: Supported Views

The following table lists the supported views and their paths.

Table 2 : Supported Views

Views	Paths
NLDM/NLPM/LVF Models	\$LIBRARY_ROOT/model/timing/lib \$LIBRARY_ROOT/model/timing/db
NLDM/NLPM/CCST/CCSP/CCSN/LVF Models	\$LIBRARY_ROOT/model/timing/ccs \$LIBRARY_ROOT/model/timing/ccs_db
AOCV Models	\$LIBRARY_ROOT/model/timing/aocv
TEST Models	\$LIBRARY_ROOT/model/test/fastscan \$LIBRARY_ROOT/model/test/tetramax
EM-IR Models	\$LIBRARY_ROOT/model/power/ansys \$LIBRARY_ROOT/model/power/voltus
HDL Functional Model	\$LIBRARY_ROOT/model/verilog
LEF	\$LIBRARY_ROOT/lef
Milkyway Database (Antenna Information i.e. CLF and edge constraints are embedded and are provided separately)	\$LIBRARY_ROOT/milkyway
NDM View	\$LIBRARY_ROOT/ndm
Physical views	\$LIBRARY_ROOT/gds \$LIBRARY_ROOT/cdl
Symbol Database	\$LIBRARY_ROOT/cdslib
Documentation	\$LIBRARY_ROOT/doc
Scripts	\$LIBRARY_ROOT/scripts



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